

Vector Space = Properties:

- ① $\vec{u} + \vec{v} \in V$ ② $\vec{u} + \vec{v} = \vec{v} + \vec{u}$ ③ $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$ ④ $\vec{u} + \vec{0} = \vec{u}$
 ⑤ $\vec{u} + (-\vec{u}) = \vec{0}$ (Addition)
 ⑥ $c\vec{u} \in V$ ⑦ $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$ ⑧ $(c+d)\vec{u} = c\vec{u} + d\vec{u}$ ⑨ $c(d\vec{u}) = (cd)\vec{u}$
 ⑩ $1\vec{u} = \vec{u}$ (Multiplication)

Note: Vector Spaces MUST include $\vec{0}$

Review: Subspaces: ① $\vec{0} \in W$ ② closed under addition ③ closed under multiplication by scalar.

Theorem: Span of any set of vectors is always a subspace

Linear transformations on Vector Spaces

onto if $\forall \vec{b} \in W \quad T(\vec{x}) = \vec{b} \quad \geq 1 \text{ soln.}$
 1-1 if $\forall \vec{b} \in W \quad T(\vec{x}) = \vec{b} \quad \leq 1 \text{ soln.}$ } $\Leftrightarrow T(\vec{x}) = \vec{0}$ only trivial

Definition: Kernel (T) = $\ker(T) = \{ \vec{x} \mid T(\vec{x}) = \vec{0} \} \Leftrightarrow \text{Null}(A)$

Defn: isomorphism: $V \xrightarrow[\text{onto}]{\text{one-to-one}} W \rightarrow$ 

if V and W are isomorphic: $\dim(V) = \dim(W)$

change of coords

$$P_B: \mathbb{R}^n \rightarrow V$$

$$P_B^{-1}: V \rightarrow \mathbb{R}^n$$

Correspondence principle: Every lin. dependence in V is accurately reproduced in $\mathbb{R}^n$ and vice versa.

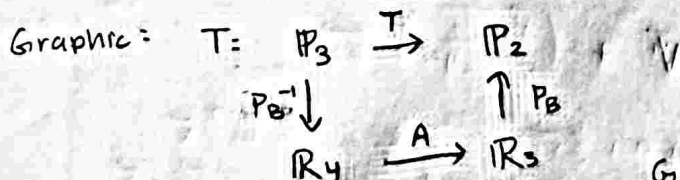
Question: does T exist $\therefore A[\vec{v}]_B = [T(\vec{v})]_C$ where $B \rightarrow$ input basis

$$\begin{aligned} \vec{v} \in \ker(T) &\Leftrightarrow T(\vec{v}) = \vec{0} \\ &\Leftrightarrow [T(\vec{v})]_C = \vec{0} \\ &\Leftrightarrow A[\vec{v}]_B = \vec{0} \\ &\Leftrightarrow [\vec{v}]_B \in \text{Null}(A) \end{aligned}$$

$C \rightarrow$ output basis.
 General Theorem:
 If $T: V \rightarrow W$ where B is basis of V
 C is basis of W

$${}_C[T]_B = \begin{bmatrix} | & | & | \\ T(\vec{b}_1) & T(\vec{b}_2) & \dots \\ | & | & | \end{bmatrix}_C$$

$${}_C[T]_B [\vec{v}]_B = [T(\vec{v})]_C$$



Generally $\text{Image}(T) = P_C \text{Col}(A)$

Rank Nullity Theorem: $\dim(V) = n$, T maps $V \rightarrow W$ $\dim(\ker(T)) + \dim(\text{Image}(T)) = n$
 from previous lec. $\leftarrow \dim(\text{Null}(A)) + \dim(\text{Col}(A)) = n$

Change of Basis

$$P_{E \leftarrow B} = \begin{bmatrix} | & | & | \\ \vec{b}_1 & \dots & \vec{b}_n \\ | & | & | \end{bmatrix}$$

$$\begin{aligned} P_{E \leftarrow B} [x]_B &= [x]_E \\ [x]_B &= [x]_E \end{aligned}$$

$$\begin{aligned} P_{C \leftarrow B} &= P_{C \leftarrow E} \circ P_{E \leftarrow B} \\ &= (P_{E \leftarrow C})^T \circ P_{E \leftarrow B} \\ &= (P_{E \leftarrow C})^T (P_{E \leftarrow B} [x]_B) \\ &= [x]_C \end{aligned}$$

In general:

$$P_{C \leftarrow B} = \begin{bmatrix} | & | & | \\ [\vec{b}_1]_C & \dots & [\vec{b}_n]_C \\ | & | & | \end{bmatrix}$$

follows

Change of basis for linear trans. T

$${}_B[T]_B = (P_{E \leftarrow B})^{-1} [T]_E (P_{E \leftarrow B})$$

Eigenvals and Eigenvecs

Fact: for diagonal $n \times n \iff \vec{e}_1 \dots \vec{e}_n$ are eigenvecs and entries = eigenvals
set of all Eigenvals forms a subspace of $\mathbb{R}^n$

Theorem: if a matrix has n distinct eigenvals, $\lambda_1, \dots, \lambda_n$, then the corresponding eigenvecs $(\vec{v}_1 \dots \vec{v}_n)$ are L.I. $\rightarrow$ Proof in Chp 15 Notes

Theorem: Dimension of an eigenspace $1 \leq \dim(E_\lambda) \leq \text{alg. multiplicity of } \lambda$
(no need to prove) geometric mult.

Remark: If $\sum_{i=1}^k \text{alg. mult } \lambda_i = n$ and for every λ_i $\dim(E_{\lambda_i}) = \text{alg mult } (\lambda_i)$
then $\sum_{i=1}^k \dim(E_{\lambda_i}) = n$.

union of bases for $E_{\lambda_1} \dots E_{\lambda_k}$ are L.I.

Theorem: Eigenvectors from different eigenspaces are L.I.

multiplicity $\rightarrow$ algebraic: power to which $(\lambda - t)^n$
 $\rightarrow$ geometric: $\dim(\text{Null}(A - \lambda I)) = \dim(\text{Eigenspace})$

Row Reduction does not preserve eigenvals (RREF $\rightarrow$ all 1s)

Similarity $A = P^{-1}DP \iff D = PAP^{-1}$

Interpret as a "change of basis"

Theorem: If A is similar to D , they have the SAME eigenvals, and each has the same geometric / algebraic mult.

Diagonalization $Q \rightarrow$ "best" basis to express a Lin. Trans.

Idea, transform $[A]$ into ${}_B[A]_B$ where B is a basis of eigenvecs.

$$\boxed{{}_B[A]_B = P^{-1} [A]_E P} \quad \text{where } P = [\vec{v}_1 \dots \vec{v}_n]$$
$$D = [\lambda_1 \dots \lambda_n]$$

Application: $A^{10} = (PDP^{-1})^{10} = PD^{10}P^{-1}$ ✓

complex eigenvals $\rightarrow$ always come in conjugate pairs $A\vec{x} = \lambda\vec{x} \rightarrow \overline{A\vec{x}} = \overline{\lambda\vec{x}} \rightarrow \overline{A}\vec{\overline{x}} = \overline{\lambda}\vec{\overline{x}}$
 $\downarrow$
 $\vec{\overline{x}}$ is also an eigenvec

Decomposition Theorem: $\vec{y} = \left(\frac{\vec{v}_1 \cdot \vec{y}}{\|\vec{v}_1\|^2}\right) \vec{v}_1 + \left(\frac{\vec{v}_2 \cdot \vec{y}}{\|\vec{v}_2\|^2}\right) \vec{v}_2 \dots$

Orthogonal Subspace Thm: If $W = \text{span}\{\vec{v}_1, \dots, \vec{v}_p\}$
 $W^\perp = \text{Null} \begin{bmatrix} \vec{v}_1^T \\ \vdots \\ \vec{v}_p^T \end{bmatrix}$
 $\vec{v}_1^T \vec{x} = \vec{v}_1 \cdot \vec{x} = 0$ for Null space

Gram-Schmidt (Inductive) Method:

$$\boxed{\vec{v}_{k+1} = \vec{x}_{k+1} - \frac{\vec{x}_{k+1} \cdot \vec{v}_1}{\|\vec{v}_1\|^2} \vec{v}_1 - \dots - \frac{\vec{x}_{k+1} \cdot \vec{v}_k}{\|\vec{v}_k\|^2} \vec{v}_k}$$

Least-Squares Algorithm. $\rightarrow$ minimizing $\|A\hat{x} - \vec{b}\|^2$

$$\boxed{A^T A \hat{x} = A^T \vec{b}} \quad \text{where } \hat{x} \text{ is the "optimal" solution}$$

Orthogonal Projection Matrix

$\text{proj}_{\vec{w}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ $[\text{proj}_{\vec{w}}] = [U U^T]$ where U is a matrix that is an ONB of $\vec{w}$.

$$\text{rank}(U U^T) = \dim(\text{Col}(U U^T)) = \dim W$$

$$\text{null}(U U^T) = \dim(W^\perp)$$

Symmetric Matrix: $A^T = A$ orthogonally diagonalizable:
 $A = P D P^{-1} = P D P^T$

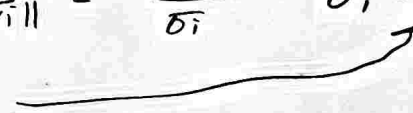
Spectral Theorem: If A is real symmetric, then

① All eigen vals of A are real ② A is orthogonally diagonalizable

Singular Value Decomp: $A = U \Sigma V^T$

$V = \begin{bmatrix} \vec{v}_1 & \dots & \vec{v}_n \end{bmatrix}$ where $\vec{v}_1 \dots \vec{v}_n$ are nonzero eigenvectors of $A^T A$

$U = \begin{bmatrix} \vec{u}_1 & \dots & \vec{u}_n \end{bmatrix}$ where $\vec{u}_i = \frac{A \vec{v}_i}{\|A \vec{v}_i\|} = \frac{A \vec{v}_i}{\sigma_i}$ $\sigma_i = \sqrt{\lambda_i}$

$\Sigma = \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \end{bmatrix}$ where $\sigma_i =$ 

Differential Eq:

1st / 2nd order LDE: use auxiliary polynomial,
 $y(t) = c_1 e^{at} + c_2 e^{bt} + \dots$ // roots are real = $a, b, \dots$

$y(t) = c_1 e^{at} (\cos(\beta t) + i \sin(\beta t)) + c_2 e^{at} (\cos(\beta t) - i \sin(\beta t))$

// roots are complex (Euler's Identity) $\alpha \pm \beta i$

Particular solution

$y_p \rightarrow$ take a "leap of faith", solve for coefficients

$y_{\text{general}} = y_{\text{part.}} + y_{\text{homogenous}}$ // superposition principle

Systems of L.D.E. // Any n^{th} order D.E can be reduced to n Linear eqs!

$T = \frac{d}{dt} - [A] \rightarrow$ find eigenvals $(\lambda_1, \dots, \lambda_n)$ of A .

soln. set = $\text{span} \{ e^{\lambda_1 t} \vec{v}_1, \dots, e^{\lambda_n t} \vec{v}_n \}$

Inner Product:

$$\textcircled{1} \langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$$

$$\textcircled{3} \langle c\vec{u}, \vec{v} \rangle = c \langle \vec{u}, \vec{v} \rangle$$

$$\textcircled{2} \langle \vec{u}, \vec{v} + \vec{w} \rangle = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle \quad \textcircled{4} \langle \vec{u}, \vec{u} \rangle \geq 0 \quad // = 0 \text{ when } \vec{u} = \vec{0}$$